Can an LLM agent do your routine command-line bioinformatics?

BioTerm-Bench v1 — 32 deterministically graded terminal tasks, 14 models, 2 infrastructures

Petr Simecek Vojtech Bystry

Central European Institute of Technology (CEITEC), Masaryk University, Czechia · simecek@mail.muni.cz

The question

Much of bioinformatics is not discovery. It is converting a GTF to a BED, checking a BAM is coordinate-sorted, normalising counts to TPM — precisely specified, tedious, easy to get subtly wrong.

So: **which parts can be delegated to an agent, and how often is the result right?**

Answering that needs a benchmark that (i) runs the tools rather than describing them, (ii) grades deterministically — no LLM judge — and (iii) runs on a laptop or an institutional node, because the data that most needs help often cannot leave the building.

What the benchmark is

32 tasks: a 25-task core plus a 7-task extension of multi-stage workflows. Each gives the agent real files, a container with the usual stack, and an exact output contract. Grading compares the produced artifact against a stored correct answer.

alignment	7	search	5
variant	7	RNA-seq	4
metadata	6		
quality control	6		

Fixed agent (Terminus-2) and harness (Harbor) across all models, so score differences reflect the model, not the scaffolding.

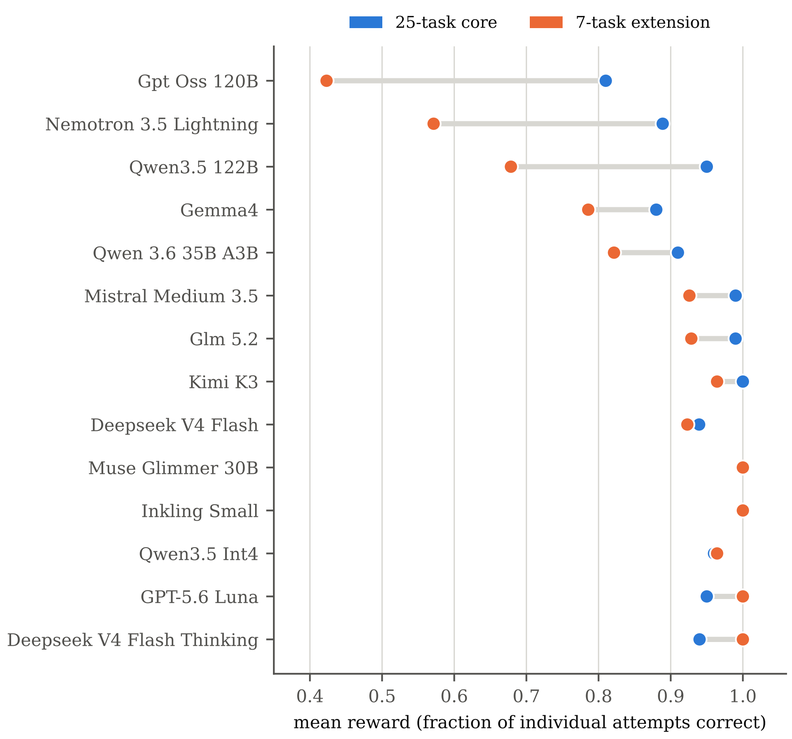
Three headline results

0.969 best mean reward on the core — **the original suite is saturated**

44% of failures are well-formed files with **wrong numbers**

9/32 tasks solved on **every** attempt by **every** model

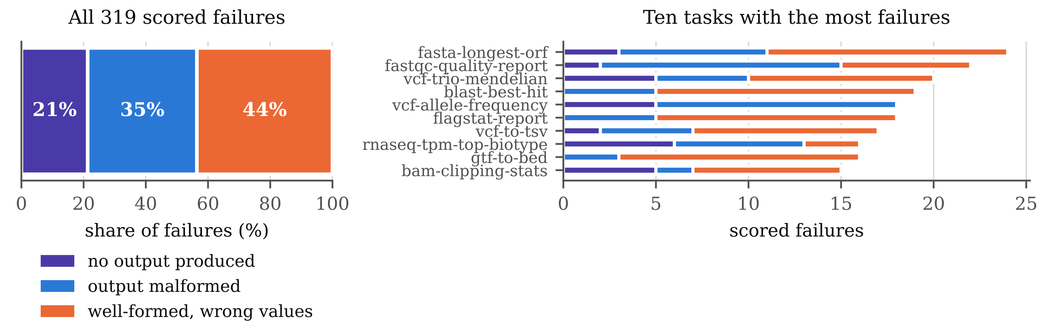
Difficulty lives in reliability, not in what a model can eventually do



Each row is one model: blue = core, orange = extension.

Best-of-four hides the gap. All five models of the OpenRouter cohort solve *every* extension task within four attempts and are therefore indistinguishable. Under the **all-four** criterion — solved on *all* attempts, the one that matters for unattended analysis — they span **2/7 to 7/7**.

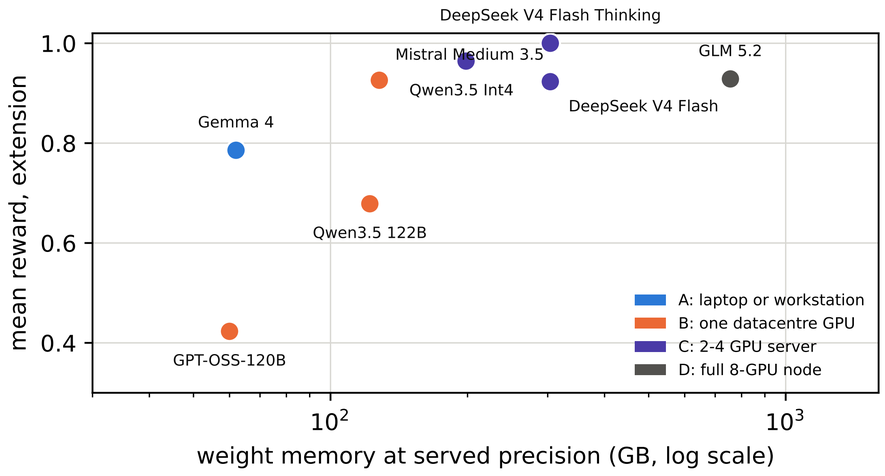
How agents fail — and why it matters



All 319 scored failures, classified from the verifiers' own reasons.

The largest category is the quiet one. A BED with correct syntax and wrong coordinates, or a flagstat summary with plausible but incorrect counts, propagates silently into whatever consumes it. Spread across ordinary tasks — `blast-best-hit`, `gtf-to-bed`, `flagstat-report` — not exotic ones.

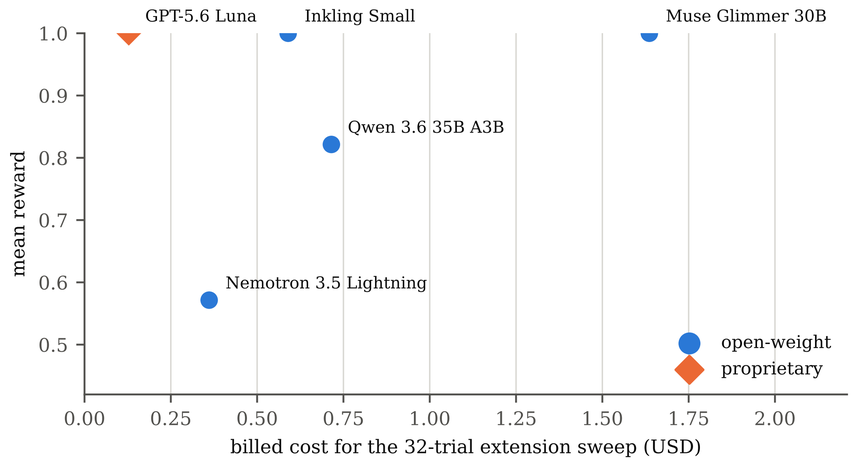
What hardware do you actually need?



Tier	Hardware	Example
A	laptop / workstation (4-bit)	Gemma 4, 62 GB
B	one datacentre GPU	Qwen3.5 122B, 122 GB
C	2–4 GPU server	DeepSeek V4 Flash, 304 GB
D	full 8-GPU node	GLM 5.2, 756 GB
E	multi-node cluster	Kimi K3, 2.8 TB

Bigger is better on average ($\rho = 0.77$) but **saturates above ∼130 GB**: a 62 GB workstation model outscores a 122 GB one needing a datacentre GPU.

Cost, openness, and what to delegate



Open weights and low price are independent. The cheapest run is also tied for the best score; the dearest costs ∼13× more for the same result.

Delegate today: format conversion and summarisation — GTF→BED, VCF→TSV, FASTA statistics, flagstat counting — with spot-checking.

Do not yet: multi-stage reasoning over biological structure — splice-junction support, trio-aware and diploid-consensus variant logic.

Rule of thumb: delegation is safe in proportion to how mechanically checkable the output is.

Take home

Routine terminal bioinformatics is largely automatable today, by models a laboratory can host itself — a workstation-class 31B model already solves most of the core suite.

Report mean reward and all-attempts, not best-of-n. Best-of-four describes a workflow with an oracle telling you which run was right. You do not have one — that is why you wanted the agent.

Check the output, not the score. When these agents fail, the most common single outcome is a plausible, well-formed file containing wrong numbers — the failure mode least likely to be caught downstream.

Freely available: `github.com/ML-Bioinfo-CEITEC/BioTerm-Bench`

Computational resources were provided by the e-INFRA CZ project (ID:90254), supported by the Ministry of Education, Youth and Sports of the Czech Republic.